TRAINING EMPLOYEES TO LOVE THEIR JOB?

A Forest-Based Analysis of Job Training and Underemployment in the UK

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QUESTION & SETUP

Recent job training → causal effect (ATE) → current underemployment?

Effect (among responders, with work-related tags) of (binary) training on the number of extra working hours desired

DML/DR/"AutoDML" forests:

→ ***D~X relationship → α***
(address regularisation bias)

→ ***ForestRiesz*** *Single GRF*
→ ***Plugin*** *D~X, Y~(D, X)*
→ ***CausalDML*** *D~X, Y~X,* $\varepsilon_Y \sim \varepsilon_D$

→ ***High-dimensionality***
→ ***Non-linearities***

UK Annual Population Survey
(Oct. 2024 - Sep. 2025)

$n = 2{,}020$ $p = 236$
$n_{treat} = 610$ $\mu_Y = 10.29$

→ ***(Treat.) selection on covariates***
→ ***Reverse/simultaneous effect***
→ ***SUTVA***
→ ***Overlap:*** $\widehat{P}(D=1|X=x_i) \in (0,1)$

→ The **empirical ForestRiesz literature** is relatively novel (Chernozhukov et al., 2021)
→ ForestRiesz's **precision** under **measured overlap** inspected

Can training interventions affect labour supply?
→ **UK Growth and Skills Levy**
→ Underemployment reflecting **productivity** or **satisfaction**?

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Training on-the-job increases productivity (multi-channel)
(De Grip & Sauermann, 2013)

Sponsored training (voluntary or not) raises retention, satisfaction
(Kosteas, 2023)

→ Higher **extra-hour premium** due to increased productivity? (Idson & Robins, 1991)

→ If your work is **fulfilling**, do you seek **more working hours**? (Golden & Wiens-Tuers, 2006)

→ If you become more productive, do you no longer "desire" **voluntary overtime** to be able to **complete** your **work**? (Beckers et al., 2008)

→ **Productivity** and job **satisfaction** (positively) **interact** (Judge et al., 2001)

OVERLAP

Inspect **D ~ X** relationship → **similarity** in **covariate distributions** across:

- ➔ Training-testing sets
- ➔ **Control-treatment sets**

If a given $\{ X_1=x_1 , \dots , X_n=x_n \}$ *appears in* ***one set****, it (approx.) appears in* ***the other*** ***(enables cross-set comparisons)***

➔ Effective sample size

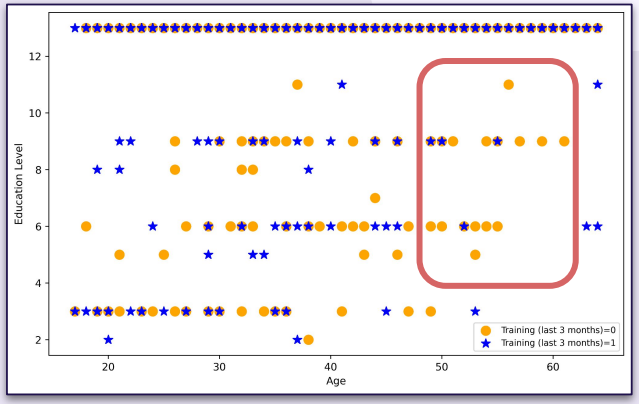
- Propensity score-based **w**
- Falls with weight disparity

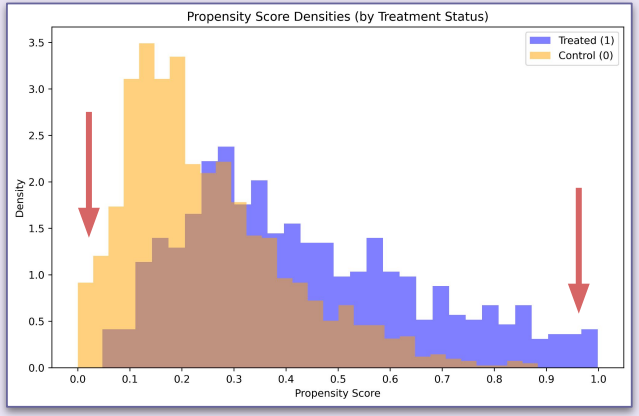
$$\left(\sum_{i=1}^{N} \widehat{w}_i\right)^2 \cdot \left(\sum_{i=1}^{N} \left(\widehat{w}_i\right)^2\right)^{-1}$$

➔ Extreme prop. scores

- (0, 0.05) U (0.95, 1)

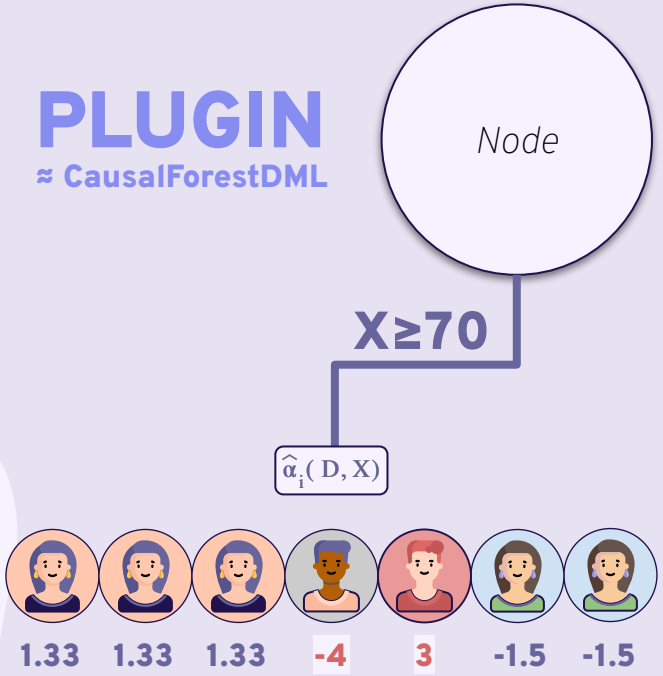
$$\widehat{P}(D=1|X=x_i)$$





PLUGIN

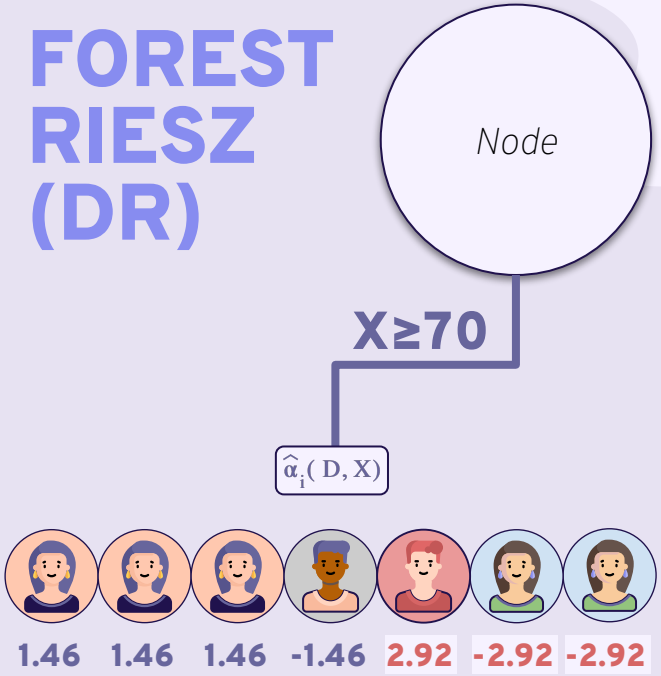
≈ CausalForestDML



$$\hat{\alpha}(D_i, X_i) = \frac{D_i}{\hat{P}(D=1|X=x_i)} - \frac{1-D_i}{1-\hat{P}(D=1|X=x_i)}$$

$$\hat{P}(D=1|X=70) = ⅓$$
$$\hat{P}(D=1|X=100) = ¾$$

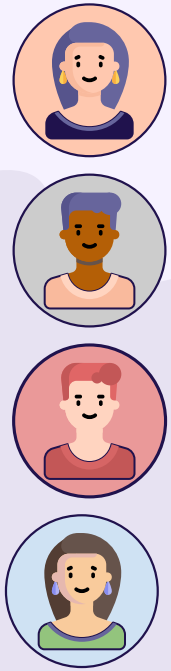
FOREST RIESZ (DR)



$$R(\alpha) = E(\alpha(D,X)^2) - 2E(\alpha(1,X) - \alpha(0,X))$$
$$\hat{\alpha}(D,X) = \operatorname{argmin}_\alpha R(\alpha)$$

Subject to **local linearity** in **D**, e.g.,

$$\alpha(D,X) = (2D-1)(mX+c)$$



X=100
D=1

X=100
D=0

X=70
D=1

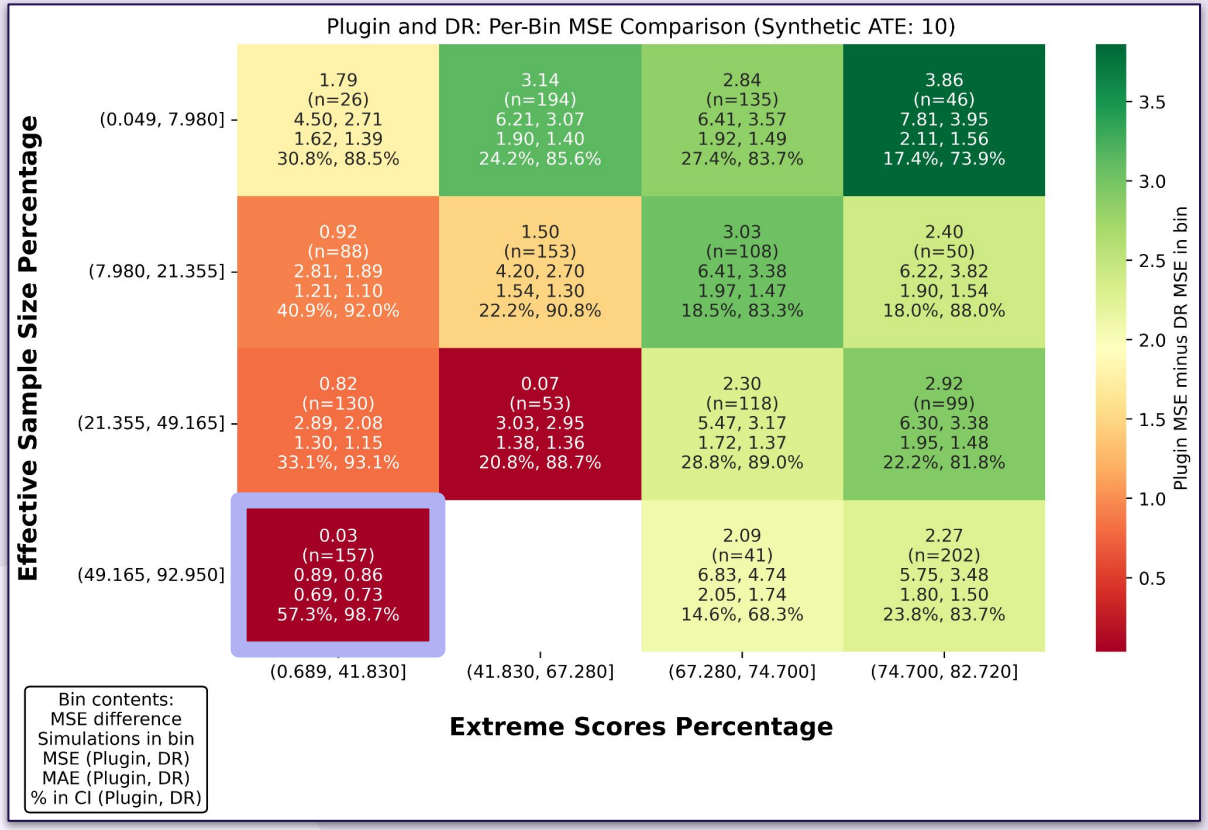
X=70
D=0

Balancing weights α, s.t.

$$E(\alpha) = 0$$
$$E(\alpha X) = E(X|D=1) - E(X|D=0)$$
$$E(\alpha D) = 1$$

$$\hat{\psi}_i = \left[\hat{\gamma}(1, X_i) - \hat{\gamma}(0, X_i)\right] + \hat{\alpha}(D_i, X_i) \cdot \left[Y_i - \hat{\gamma}(D_i, X_i)\right]$$

SIMULATION EXERCISE



→ Fixed (real) covariates
→ Treatment **selection:** on **random covariates,** with **random intensity**
→ Non-linear & **random** mean-zero **het. effects**

Plugin *suffers from* ***bias***
→ ***Overfitting****: no cross-fit.*
→ ***Extrapolation****: hinted by enlarged difference under* ***weight disparity*** *or* ***extremeness*** → **α** *issue*

ForestRiesz may bolster precision (rel. to Plugin)
→ **MSE** (overall precision)
→ **Coverage**
→ Useful **general insights**

EMPIRICAL RESULTS

Forest Riesz (DR) (CF; scaled X)	Forest Riesz (DR) (CF; unscaled X)	Forest Riesz (DR) (no CF; scaled X)	Forest Riesz (DR) (no CF; unscaled X)	CausalForestDML (CF, scaled X)
β_{hat} = -0.169 std. err. = (0.838)	-0.169 (0.835)	-0.267 (0.310)	-0.268 (0.310)	0.156 (0.456)
-0.789 (1.323)	-0.794 (1.316)	**-0.985**** **(0.399)**	**-0.986**** **(0.400)**	*GATE: part-time workers (n=1061)*
0.528 (1.113)	0.533 (1.112)	0.528 (0.481)	0.526 (0.481)	*GATE: full-time workers (n=959)*
-0.009 (0.973)	-0.009 (0.970)	-0.075 (0.330)	-0.077 (0.331)	*GATE: not seeking other work (n=1612)*
-0.796 (1.579)	-0.799 (1.578)	-1.023 (0.810)	-1.023 (0.811)	*GATE: seeking other work (n=408)*

REFLECTION & FUTURE WORK

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The observed ATE is often negative, yet insignificant
→ GATEs similarly questionable

Biggest challenges:
→ Grasping Riesz representers and the overlap "research gap"
→ Depth over breadth (literature)

→ **Scaling** practically **irrelevant** (as expected in a forest)
→ **Non-cross-fitted** outcomes may suffer from **overfitting bias**

What worked best:
→ **Robustness** (**sanity**) checks: **simulations**, **scaling**, **cross-fit**.
→ Looking at researchers' code!

→ **Productivity** and **satisfaction** facets (for- and against- extra hours) captured → **cancel out**?
→ **Underemployment** as a relatively **indep. decision**?

Forest Riesz-CausalForestDML:
→ Deep dive into **architectures**?
→ **Simulations** between them?
→ Exploring **RieszNet**?
→ GATEs?